

AI-Driven Analysis of Political Bias in Financial Media Impact on Retail Investor Behavior and Stock Market Dynamics



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The Disconnect Retail vs. Institutional Realities

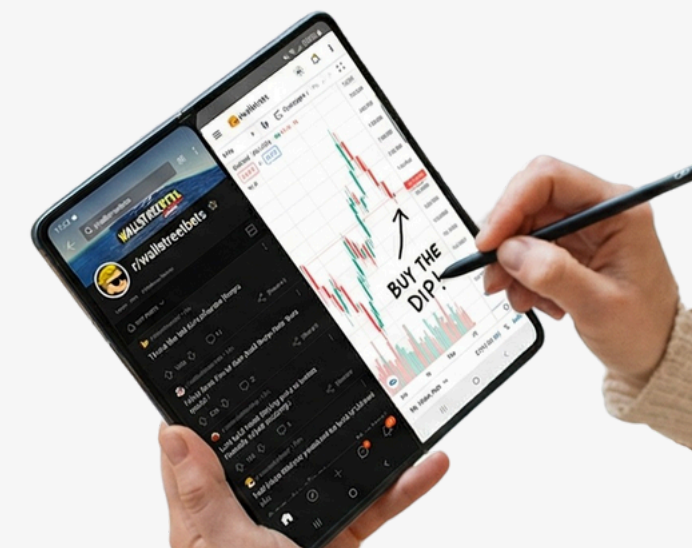
Algorithmic Trading Blind Spots in a
Polarized Economy

The Shift. Retail investors (Reddit/WSB) do not trade on fundamentals; they trade on narratives and viral sentiment".

The Gap. Current models miss the "Reputational Prior" how the source of the news filters the interpretation of the risk.

The Reality. The "Attention Economy" incentivizes polarization.

The Problem. Traditional algorithms treat news providers as neutral conduits of information.



Does Ideological Framing Quantifiably Improve Prediction?

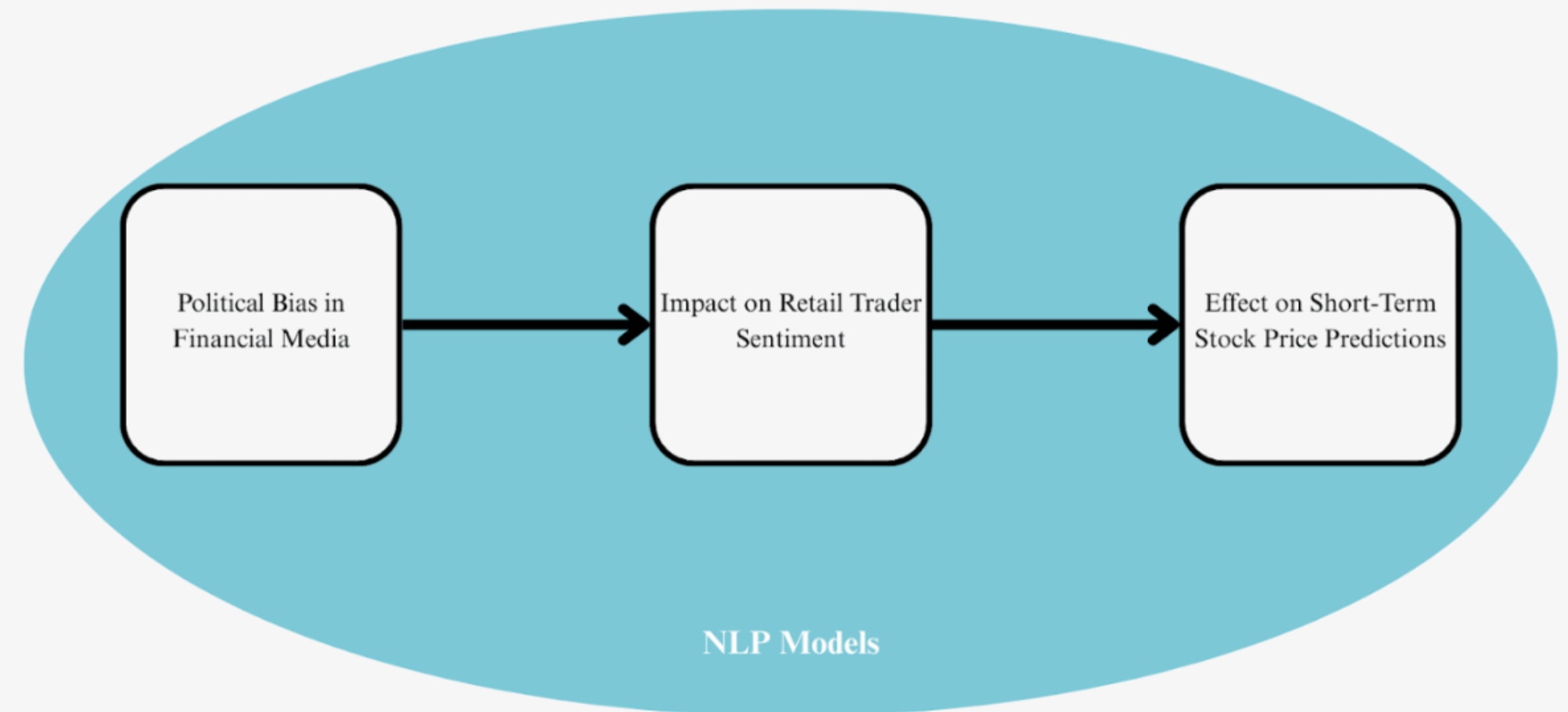
Research Question.

"How does the influence of political bias in major financial news outlets on retail trader sentiment improve short-term stock price predictions through NLP models?"

Objective. To determine if political bias is "noise" to be ignored or a "risk factor" to be captured.

Scope. 10 High-Volatility Tickers (S&P 500/NASDAQ), June 2024–June 2025

Moving Beyond Efficient Markets to Adaptive Narratives



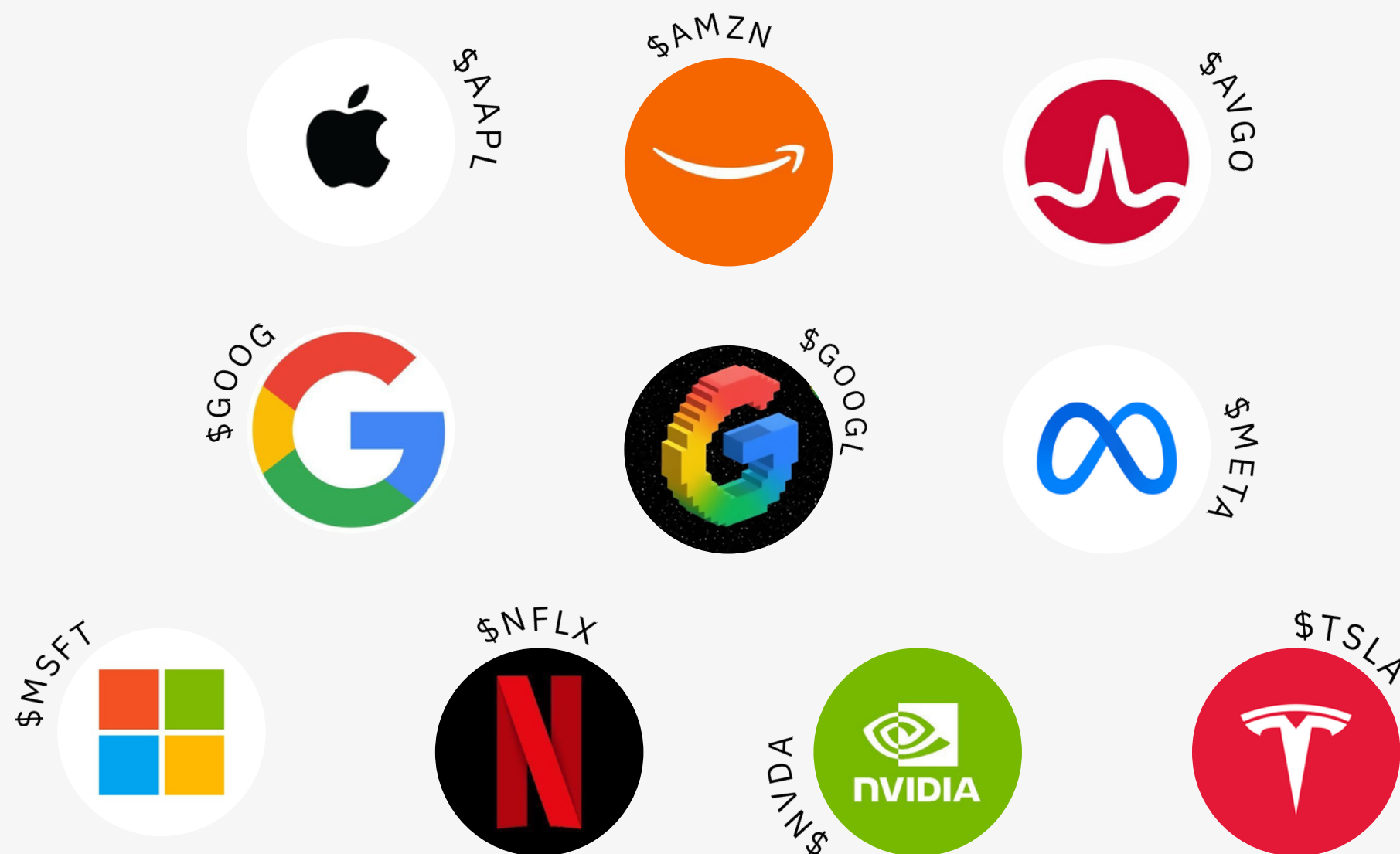
Noise Trader Theory (Black, 1986). Retail sentiment creates temporary mispricing (noise) that arbitrageurs hesitate to correct.

Reputational Prior. We model the publisher's brand (bias) as a filter for information processing, not just the text itself.

Data Architecture

Constructing a Multi-Source Behavioral Dataset

Market Data. 10 High-beta tickers. **Target:** Next-Day Direction (Binary)



2,570 Ticker-Day observations

Sample Window. June 2024 – June 2025.

Professional News.

8,830 articles

 **GNews**



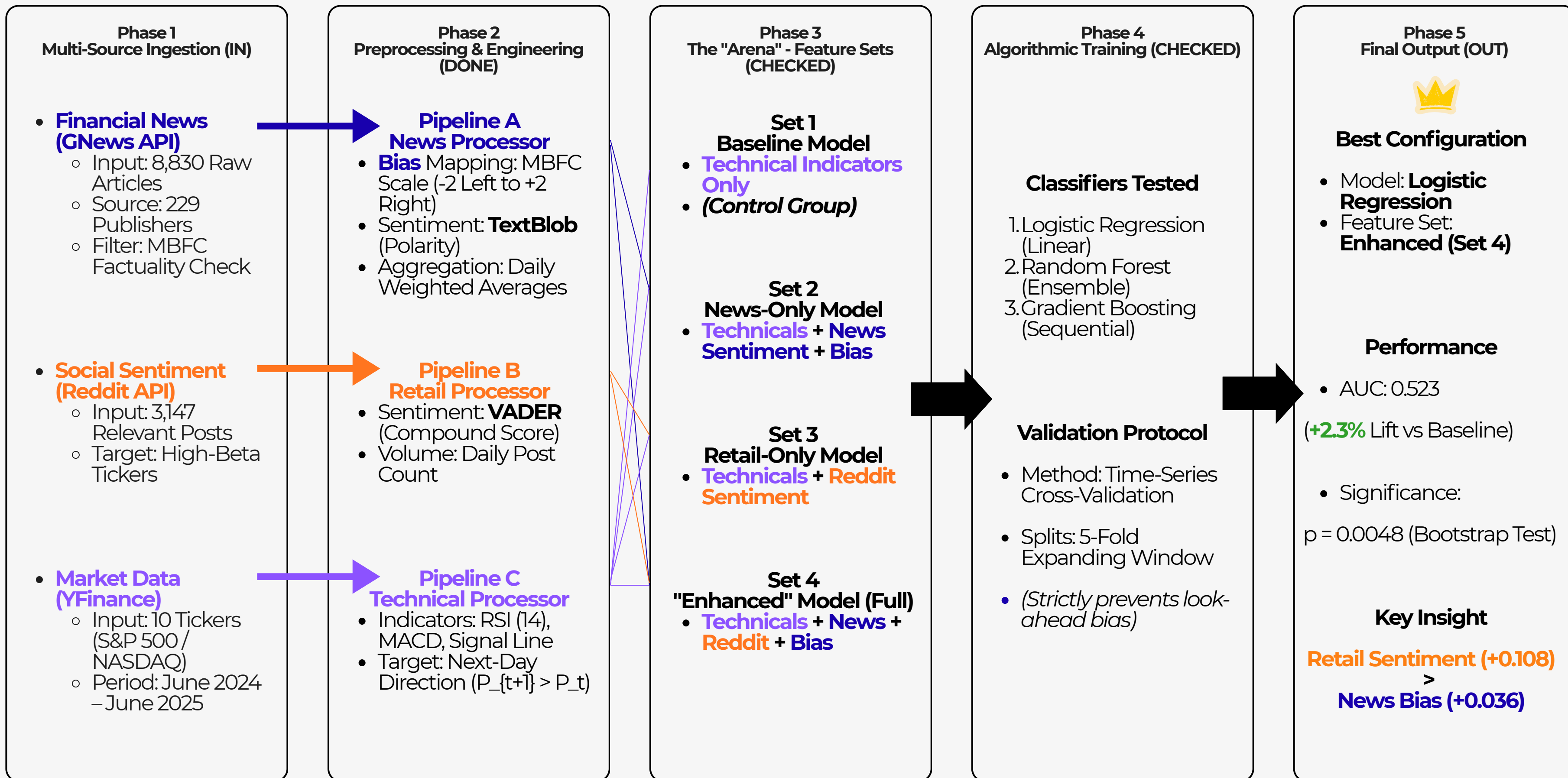
TextBlob used for lexicon-based scoring.

Retail Sentiment.

3,147 reddit posts



Aggregated volume and sentiment via VADER



Isolating the Signal from the Noise

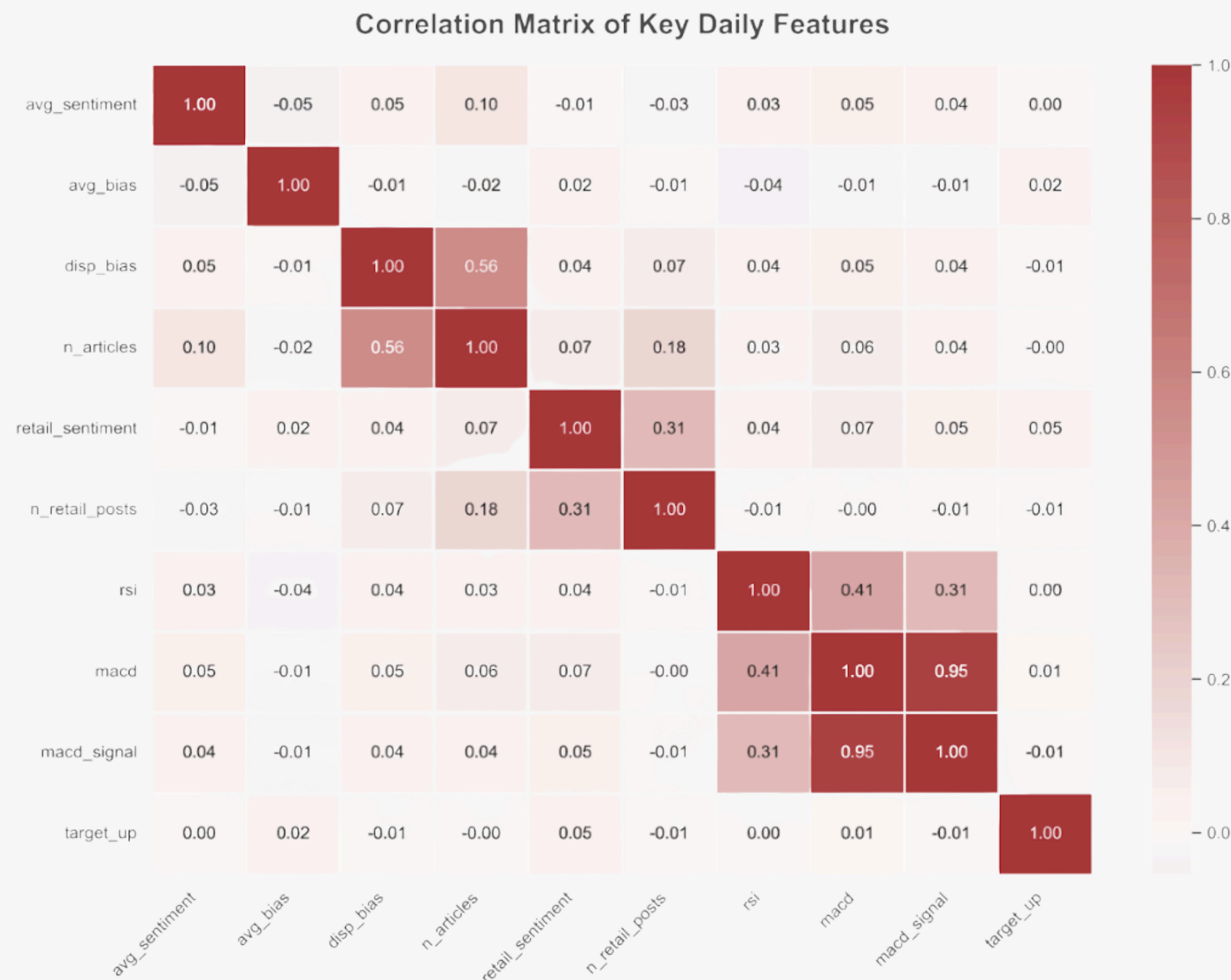
Feature Engineering Strategy

• Validation

The heatmap proves that our textual features (Sentiment/Bias) are **not multicollinear** with technical indicators (MACD/RSI).

• Novel Features

- **avg_bias:** The directional slant of the news coverage.
- **disp_bias:** The disagreement (standard deviation) among publishers.

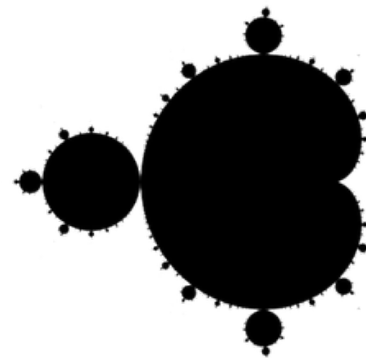


NLP Strategy: Tuning the Tool to the Text

Why the split of Sentiment Engines

Target.

8,830 Financial
News Articles
(GNews).



TextBlob

The "Formal" Engine

For Professional News

What is it.

A general-purpose
lexicon-based library.

How it works.

Assigns
polarity (-1 to +1)
based on
adjectives and
grammatical
structure.

Why here.

Financial news follows strict
grammatical rules. TextBlob excels
where tone is subtle.

Target.

3,147 Reddit Posts

How it works.

Specifically tuned for
social media contexts.



The "Social" Engine

For Retail Chatter

What is it.

Valence
Aware
Dictionary
and
sEntiment
Reasoner.

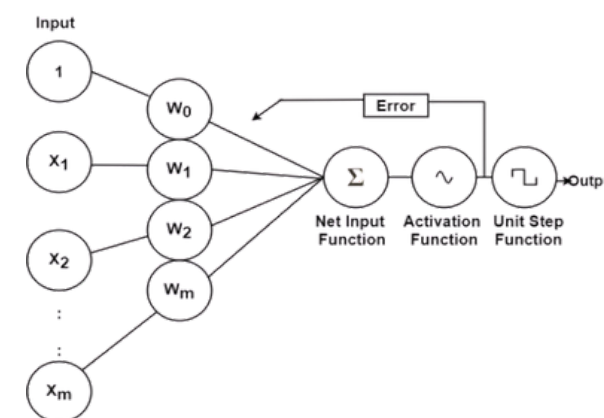
Why here.

- Capitalization: "BUY" > "buy"
- Punctuation: "Great!!!" > "Great"
- Emojis: Handles 🚀, 💎, 📈 directly.
- Slang: Recognizes "FOMO", "YOLO", "HODL".

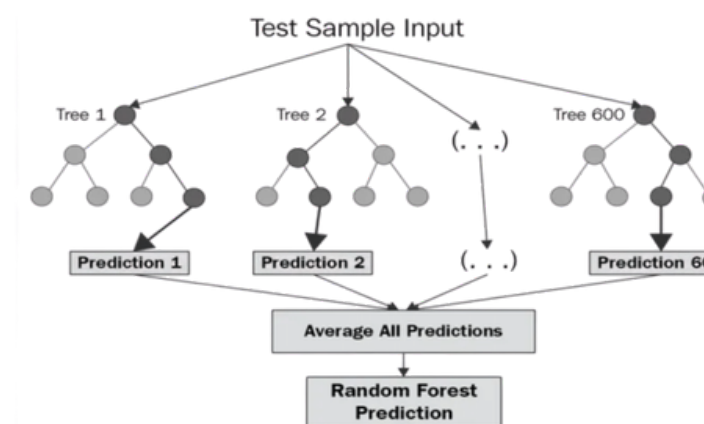
Modeling & Validation

A Rigorous Time-Series Approach

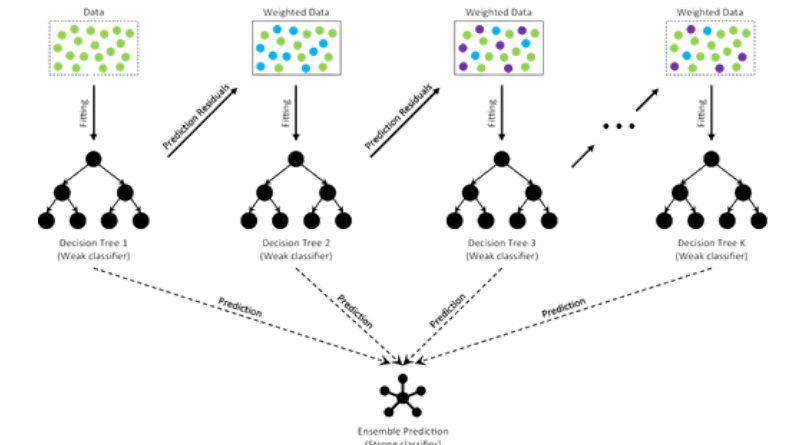
- Models:
 - Logistic Regression
 - Random Forest (Ensemble)
 - Gradient Boosting (Boosting)



(Biswas, Sandipan & Ghosh, Shrivath & Roy, Sandip & Bose, Rajesh & Soni, Sanjay, 2023).



(Amanoui, Sandy & Abdulazeez, Adnan & Zeebaree, Diyar & Ahmed, Falah, 2021).

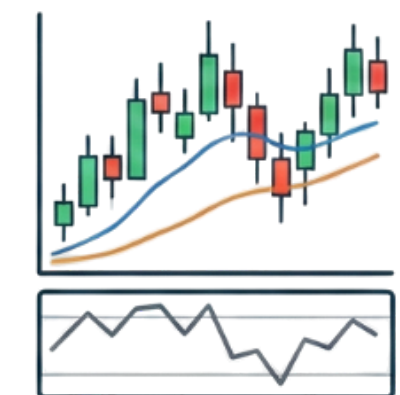


(Deng, Haowen & Zhou, Youyou & Wang, Lin & Zhang, Cheng, 2021).

- Validation:
 - Strict Time-Series Cross-Validation (5 Splits)



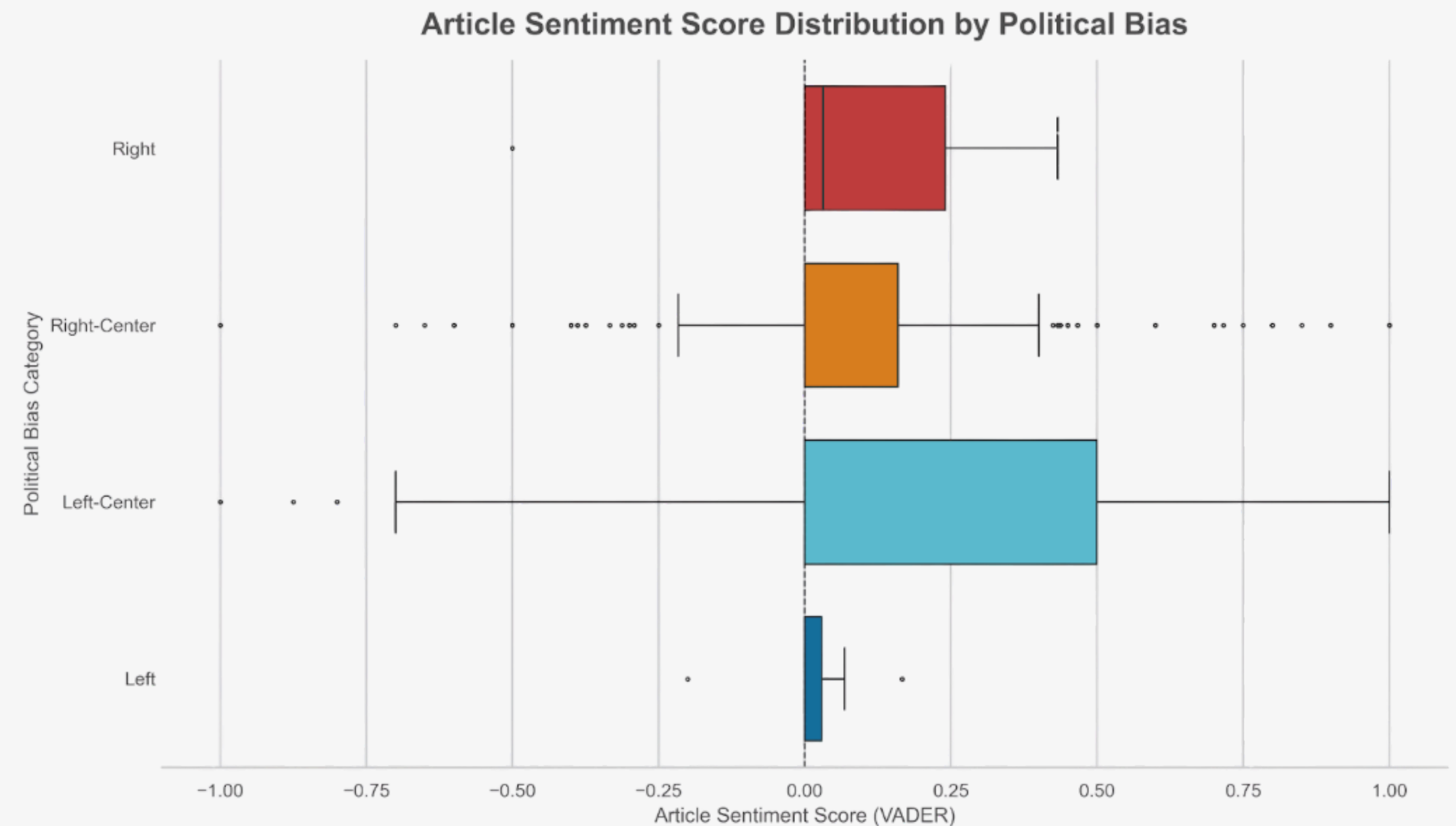
- Baselines:
 - "Naive" strategies (Always Up, Persistence)
 - "Technical Only" baseline.



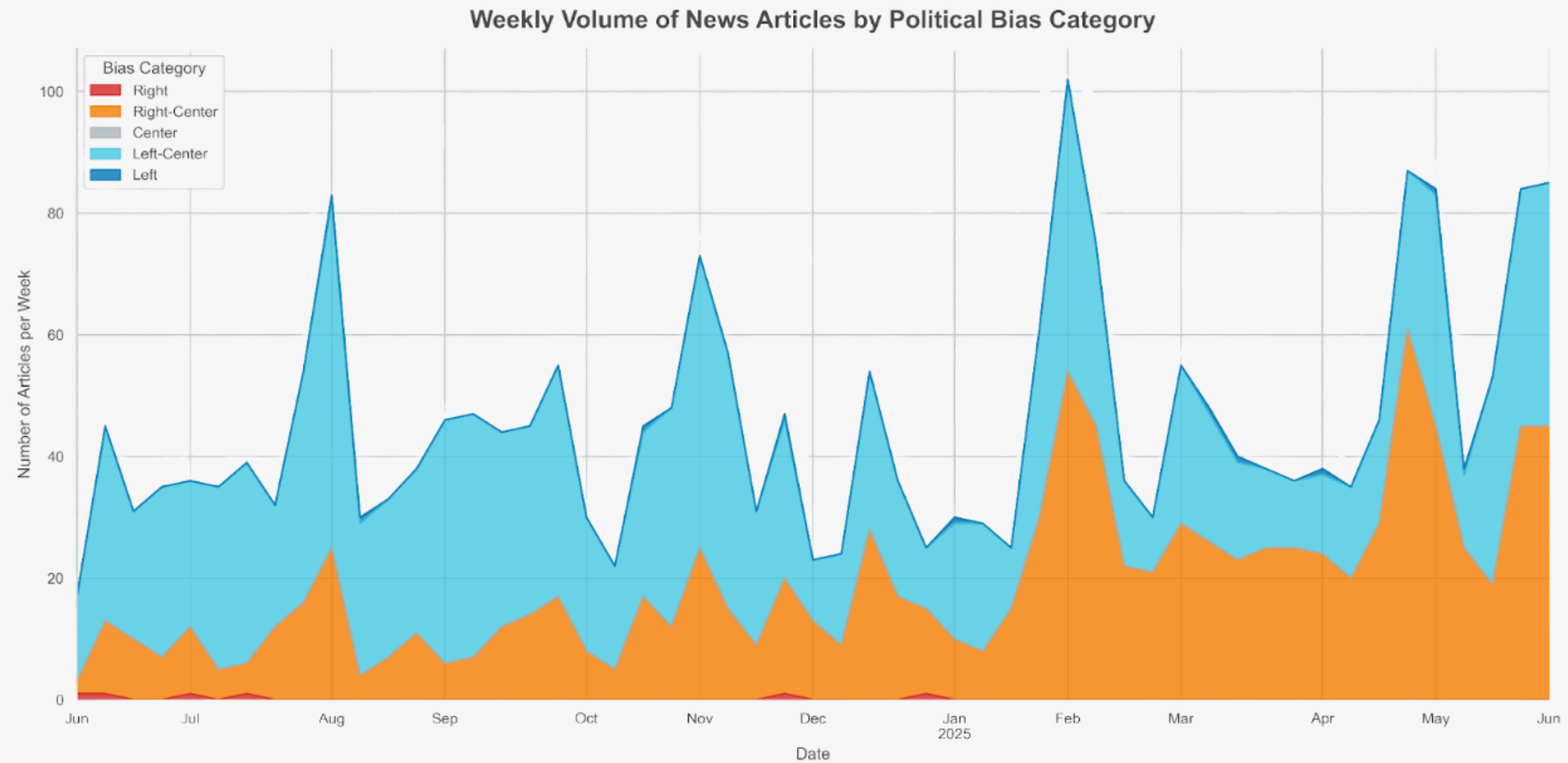
Descriptive Analysis The Polarization Gap

Sentiment is Not Distributed Evenly

- Structural sentiment gap.
- "Left-Center" sources are structurally more optimistic (Median 0.500 upper quartile) than "Right-Center" sources.
- "Bias" is not just about political stance, but about emotional framing.



Descriptive Analysis



Behavioral Features deliver a Consistent Predictive Lift

Predictive Performance

- **The Lift**

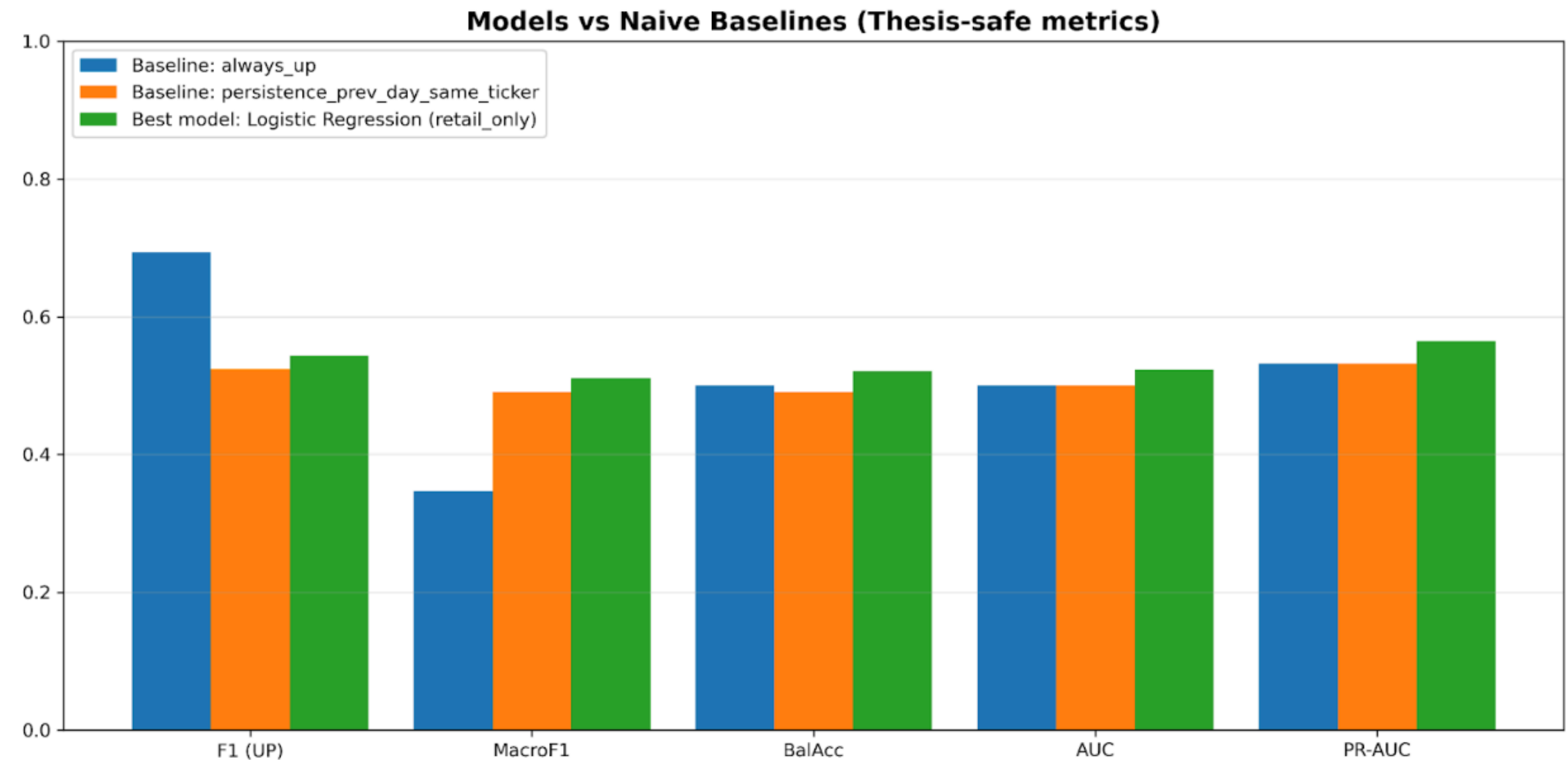
- The "Retail Only" and "Enhanced" models consistently outperform Naive benchmarks.

- **The Numbers**

- AUC of **0.523** (+0.023 vs Baseline).
- While the number seems small, it is statistically significant (**$p < 0.05$**).

- **Robustness**

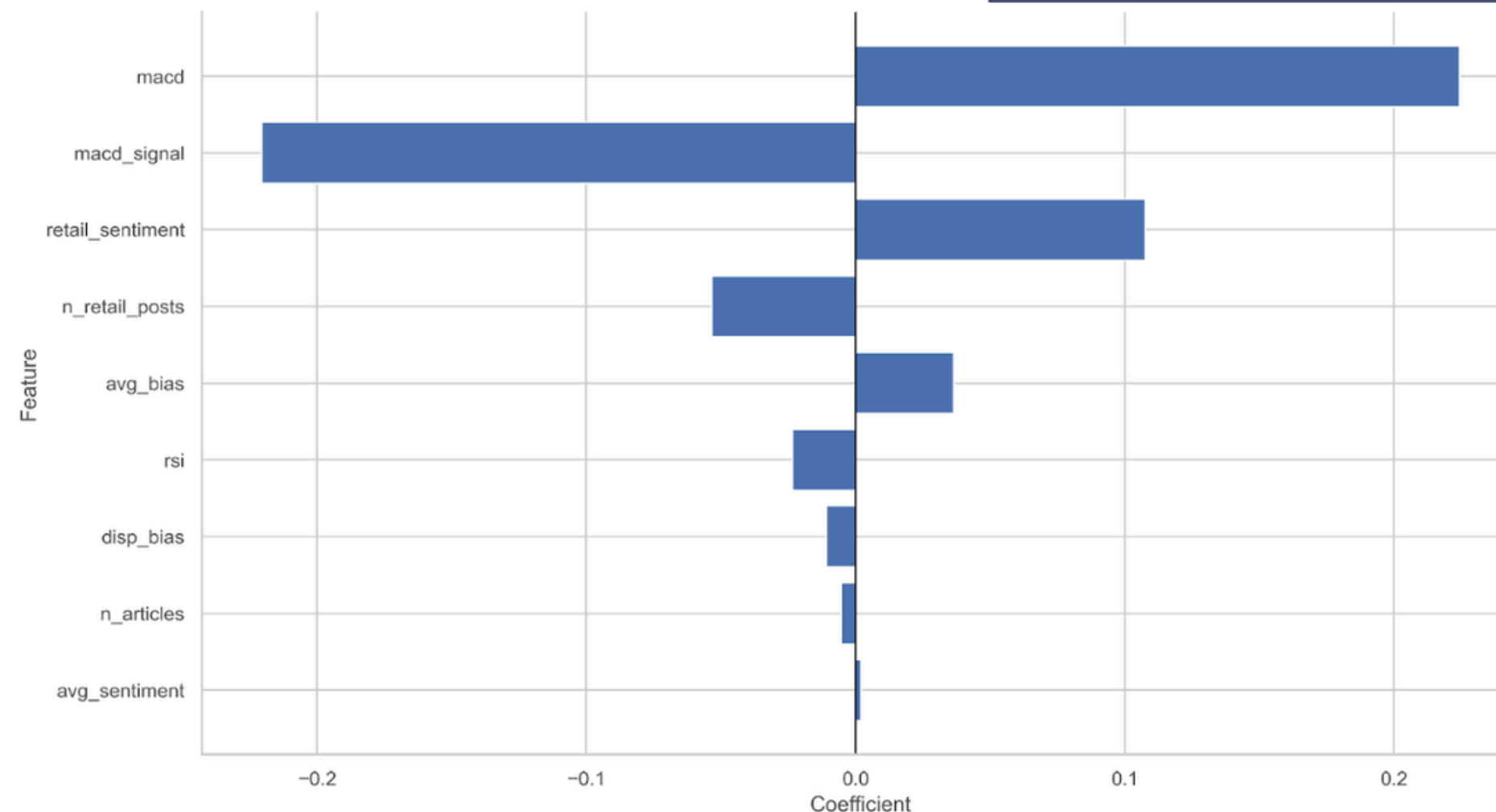
- This lift holds across different model families (Linear and Non-Linear).



The Lift: +2.3% improvement.

Retail Noise Overpowers Professional News

Feature	Coefficient
macd	0.224
macd_signal	-0.221
retail_sentiment	0.108
n_retail_posts	-0.053
avg_bias	0.036
rsi	-0.023
disp_bias	-0.011
avg_sentiment	0.002



- **Dominance**

- **retail_sentiment (0.108)** far outweighing **avg_sentiment (0.002)**.

- **Bias Matters**

- **avg_bias (0.036)**
- who reports the news matters more than what the sentiment score is.

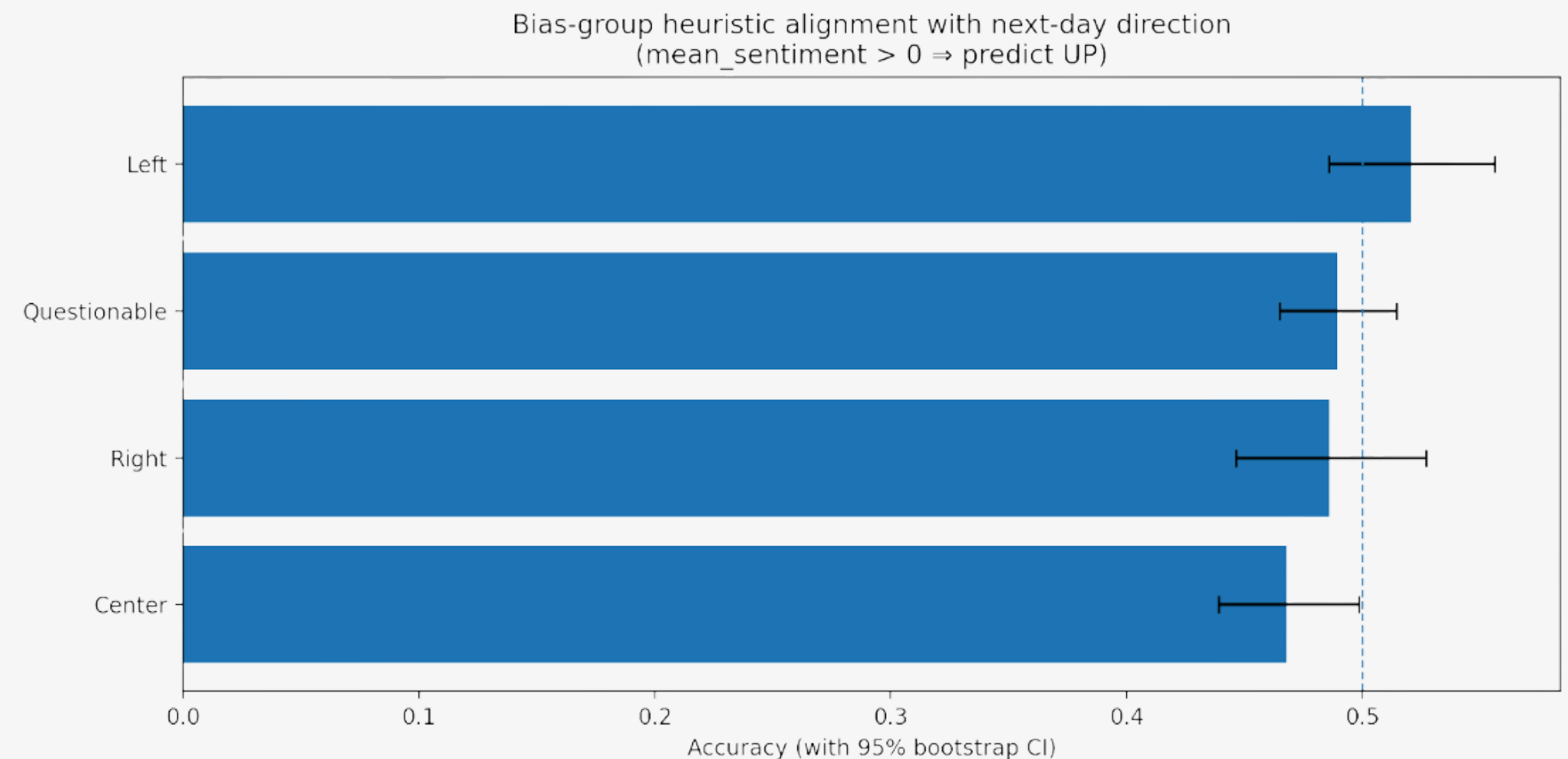
- **Consensus over Conflict**

- **disp_bias (disagreement)** was negligible. The market follows the dominant narrative, not the debate.

*The crowd's reaction (+0.108) is **50x** more powerful than the news sentiment itself (+0.002).*

Ideological Direction Impact on Accuracy

The "Left-Leaning" Anomaly



Finding. Left-leaning outlets had a higher directional accuracy (52.1%) than Centrist outlets (46.8%) during this sample period.

Implication. Neutrality does not equal predictiveness. The market may be reacting more strongly to "framed" narratives than balanced reporting.

The "Decoupling" of News and Retail

Interaction Effects

Interpretation.

In viral/meme stock events, the "crowd" decouples from the institutional news cycle.

Regime Dependency.

Bias signals work best in "quiet" regimes; Retail signals work best in "viral" regimes.

The Phenomenon.

When retail volume(**Reddit posts**) spikes (>2 SD), the predictive weight of professional news bias fades.

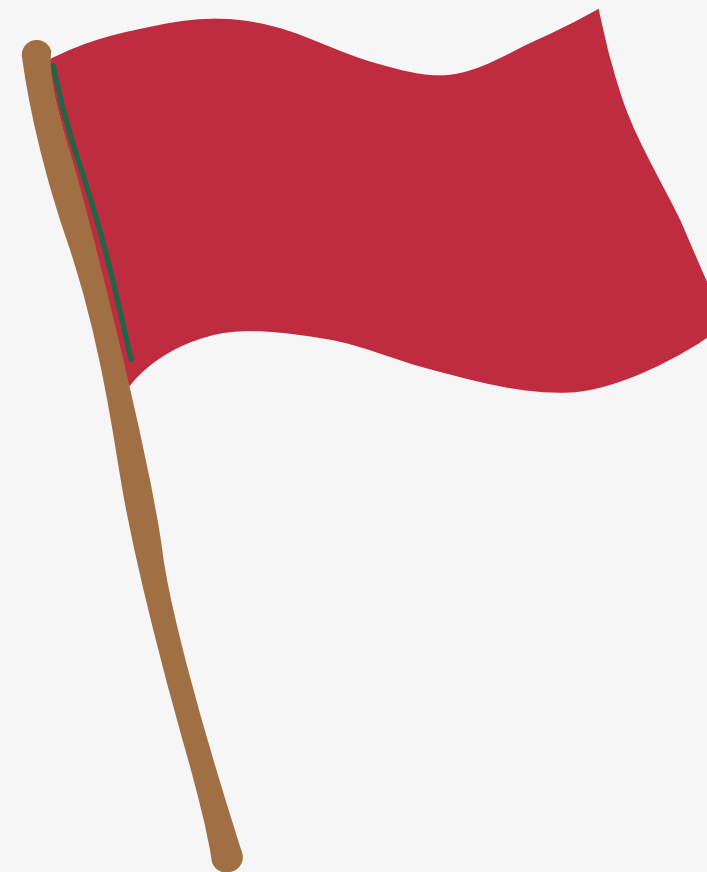


A "Narrative Risk Premium" in Tech

Sector Specifics

Implication.

The 0.023 AUC lift is likely a "Narrative Risk Premium" specific to high-volatility, culturally relevant stocks.



This is not a universal law for utilities or low-beta stocks, but a specific mechanic of the "Attention Economy".

Context.

The model works best on high-attention tickers (TSLA, NVDA).

Action.

If News is Negative but Retail is Euphoric ($Z\text{-score} > 2.0$),
flag for high volatility reversal risk.

Risk Management.

Do **not** use this as
a standalone
trading bot (Alpha
decay/Costs).

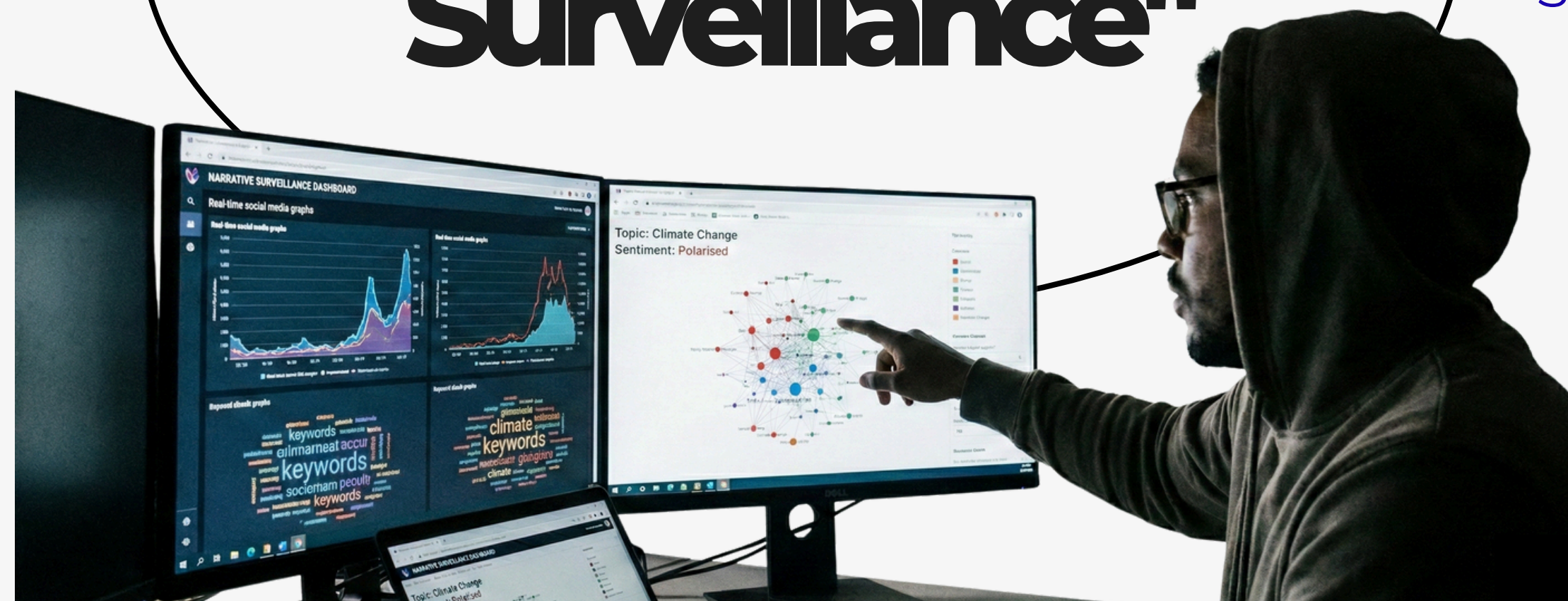
Use it as a Risk
Filter

Managerial Implications (Investors)

Operationalizing "Narrative Surveillance"

Strategy.

Monitor the
divergence
between
**Institutional
Sentiment (News)**
and **Retail
Sentiment
(Reddit)**





The Loop.

If algorithms trade on bias (which is profitable), they incentivize publishers to be **more biased** to capture algorithmic attention.

Managerial Implications (Platforms & Regulators)

The Ethics of Algorithmic Amplification

Recommendation.

Regulators need to monitor "**Narrative Anomalies**" where news bias deviates sharply from economic fundamentals.

Constraints of Real-World Deployment

Limitations

Latency.

The signal is real, but high-frequency execution requires overcoming latency. A 5-second delay could erode the 52% accuracy edge.

Transaction Costs.

The lift is incremental. Execution costs (slippage/fees) must be factored in; this is not "easy money"

Sample Window.

2024-2025 is a specific political regime (Election cycle).

The "Bias" signal might decay in technocratic years

The Market as a Voting Machine for Narratives

Final Conclusion

Yes, it works.

Political Bias and
Retail Sentiment
improve prediction
(+2.3% Lift)

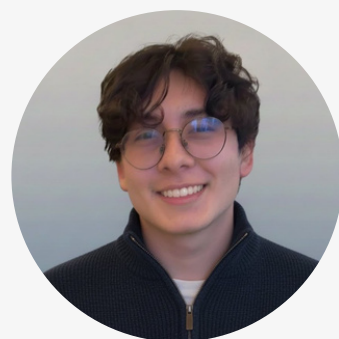
Hierarchy.

Retail Sentiment >
News Bias > News
Sentiment. The
"Crowd" transmits
the volatility"

Contribution.

Provided a
transparent, open-
source framework
to quantify
"Reputational Priors"
without using black-
box LLMs

thank you



LinkedIn

Q&A



GitHub